Féidearthachtaí as Cuimse
Infinite Possibilities

Machine Learning

Lecture 5: Evaluation

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Slides adapted from Sarah Jane Delany and book slides from: Fundamentals of Machine Learning for Predictive Data Analytics. Kelleher, Mac Namee and D'Arcy

Overview

- The idea behind decision trees
- Shannon’s entropy model
- Splitting criteria in decision trees
- The ID3 algorithm in decision trees
- Handling various feature types in decision trees
- Improving decision tree training

Introduction

- Evaluation is a critical task in machine learning.
- Evaluation is used to
  - Estimate how a model will perform when deployed.
  - Compare the performance of different models on a particular task
- Evaluation experiments generally follow these steps:
  - Decide on how to measure performance (ie. evaluation measures to use)
  - Run the model one or more times on a dataset or a collection of datasets.
  - Compare the performance of the model with existing benchmark models using the evaluation measure(s)
- The exact experimental setup depends on the hypothesis that is being tested

Hypothesis testing

- The goal of hypothesis testing: formally examine two opposing hypotheses, H_0 and H_A. These two hypotheses are mutually exclusive, so one is true to the exclusion of the other.
- Null Hypothesis H_0: States the assumption to be tested. e.g. There is no difference between the performance of two machine learning algorithms.

Hypothesis testing

- In ML, we often talk about hypotheses at different levels,
- High-level
  - e.g. "*A Neural Network will perform better than Naive Bayes on this particular problem*"
- Low-level (the output from a model)
  - e.g. "*This email is spam*"

Type I and Type II errors

- *Type I error*: Rejecting H_0 when it is, in fact true.
  - i.e. “false positive” - detecting a difference, when none exists.
- *Type II error*: Failing to reject H_0 when it is in fact false.
  - i.e. "false negative" - concluding there is no difference, when there is.

Statistical Test Result

Real World	H_0 Rejected	H_0 Not Rejected
There is a real difference	**Correct** A Hit	**Type II Error** Missed a real difference
There is in fact no difference	**Type I Error** False alarm	**Correct** Right to be sceptical of H_A

Type I and Type II errors

- Depending on the hypothesis being tested, these errors will have different costs associated with them.

Null Hypothesis	**Type I Error "False Positive"**	**Type II Error "False Negative"**
"Person X does not have the virus"	X is clear but marked as infected.	X is infected but marked as well.
Cost of the error	X is required to isolate for 14 days.	X can infect others.

Null Hypothesis	**Type I Error "False Positive"**	**Type II Error "False Negative"**
"Person X is not guilty of the crime"	X is found guilty of a crime, when they are actually innocent.	X is found not-guilty of a crime, when they did actually commit the crime.
Cost of the error	Social cost of sending an innocent person to prison.	Risk of letting a guilty criminal go free.

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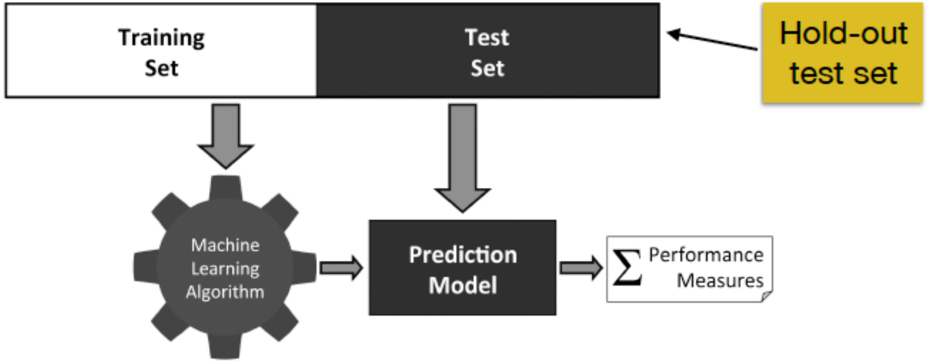
Standard Approach to Evaluation

- Use a labelled test set of known predictions and compare the actual labels with the predictions to measure performance.



- A mandatory requirement is that the test data must be separate from the training data.
  – Using the same training data for testing can produce unrealistic accuracy results that are "too good to be true".
- Performance on test data estimates how the model can generalise beyond the data used to train it.

Measuring performance

- No model will ever be perfect
- Important to consider the performance required by the task.
  - Medical diagnosis → needs highly accurate diagnosis.
  - Predicting customer response to an ad → anything better than random will deliver profit.
- Must align the evaluation measures with the modelling task

Classification Accuracy

- The simplest measure of classification is the Misclassification Rate or Error Rate.

$$\text{MR} = \frac{\#\text{incorrect predictions}}{\#\text{predictions}}$$

- The associated measure is Accuracy.

$$\text{ACC} = \frac{\#\text{correct predictions}}{\#\text{predictions}}$$

Test Set

ID	Target	Pred.
1	spam	ham
2	spam	ham
3	ham	ham
4	spam	spam
5	ham	ham
6	spam	spam
7	ham	ham
8	spam	spam
9	spam	spam
10	spam	spam

$$\text{MR} = \frac{2}{10} = .2$$

$$\text{ACC} = \frac{8}{10} = .8$$

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Confusion Matrix

- The confusion matrix summarises the performance of a classifier when compared with the actual target classes (“ground truth”).

		Predicted Class	
		Pos	**Neg**
Actual Class	**Pos**	**TP** **True Positive** Correct!	**FN** **False Negative** (Type II error)
	Neg	**FP** **False Positive** (Type I error)	**TN** **True Negative** Correct!

- **Clinical Example**: Predict a case as positive (person has the disease) or negative (person does not have the disease)
  - **TP** = Sick people correctly predicted as sick.
  - **FP** = Healthy people incorrectly predicted as sick.
  - **TN** = Healthy people correctly predicted as healthy.
  - **FN** = Sick people incorrectly predicted as healthy.

Confusion Matrix

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

		Prediction	
		spam	**ham**
Actual	**spam**	6	3
	ham	2	9

Classification Evaluation Measures

- Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

- Misclassification Rate

$$\text{MR} = \frac{FP + FN}{TP + FP + TN + FN}$$

- True Positive Rate: Focus on TPs aka *Sensitivity & Positive Recall*

$$\text{TPRate} = \frac{TP}{TP + FN}$$

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

- False Positive Rate: Focus on FPs

$$\text{FPRate} = \frac{FP}{FP + TN}$$

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

- True Negative Rate: Focus on TNs Also called *Specificity & Negative Recall*

$$\text{TNRate} = \frac{TN}{FP + TN}$$

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

Confusion Matrix

ID	Target	Pred.	Outcome
1	spam	ham	FN
2	spam	ham	FN
3	ham	ham	TN
4	spam	spam	TP
5	ham	ham	TN
6	spam	spam	TP
7	ham	ham	TN
8	spam	spam	TP
9	spam	spam	TP
10	spam	spam	TP

ID	Target	Pred.	Outcome
11	ham	ham	TN
12	spam	ham	FN
13	ham	ham	TN
14	ham	ham	TN
15	ham	ham	TN
16	ham	ham	TN
17	ham	spam	FP
18	spam	spam	TP
19	ham	ham	TN
20	ham	spam	FP

		Prediction	
		spam	**ham**
Actual	**spam**	6	3
	ham	2	9

Assuming spam is the positive class

Accuracy = .75

Error Rate = .25

TPRate = .66

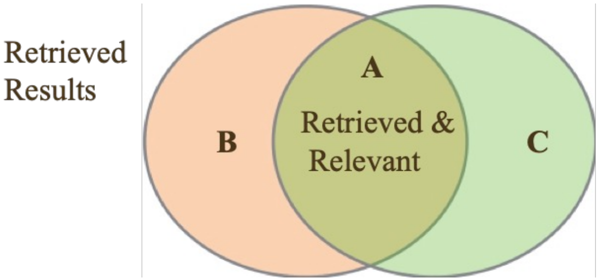
TNRate = .81

FPRate = .79

FNRate = .33

Precision and Recall

- These are measures that come originally from the Information Retrieval domain.
- Precision: proportion of retrieved results that are relevant.
- Recall: proportion of relevant results that are retrieved.



$$\text{Precision} = \frac{A}{A+B}$$

$$\text{Recall} = \frac{A}{A+C}$$

Search Example: Given a collection of 100k documents, we want to find all documents on "hospital waiting lists". In fact, 45 relevant documents actually exist. Perform a search, 10 docs were returned, 9 of which are relevant documents.
➡️ Precision = 9/10 = 90% of retrieved results were relevant.
➡️ Recall = 9/45 = 20% of all possible relevant results were retrieved
➡️ P@10 used in Web Search, how many relevant webpages were returned in top 10

Recall and Precision

- Applying these to ML, these are *class* measures.
- Recall is class accuracy, see TPRate and TNRate.
- Recall is the proportion of the actual instances of the class that are correctly predicted.

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

Recall

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

Negative Recall

- Precision - the proportion of the predicted instances of the class that are correct

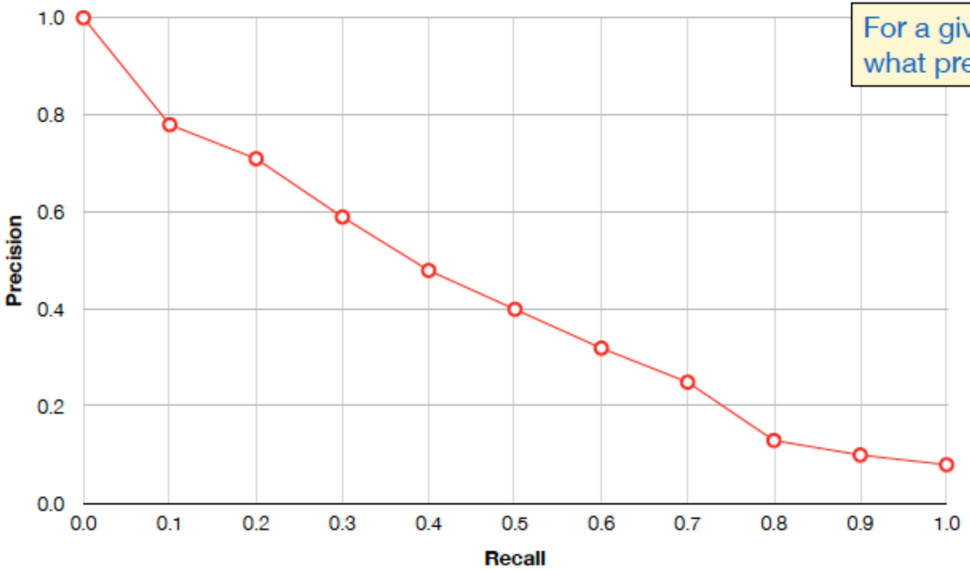
$$\text{Precision} = \frac{TP}{TP + FP}$$

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

$$\text{Negative Precision} = \frac{TN}{TN + FN}$$

		Predicted	
		Pos	Neg
Actuals	Pos	TP	FN
	Neg	FP	TN

P/R Curve



For a given level of recall,
what precision do we achieve?

- Using a Precision Recall (PR) curve, plot the trade-off between the two measures.
- Used to study the output of a binary classifier.
- Measure precision at fixed recall intervals.
- Improving one will dis-improve the other.

Precision vs. Recall

These are *class* measures:

Spam Recall = 4/(4+2) = .66

Non Spam Recall = 3/(1+3) = .75

Spam Precision = 4/(1+4) = .8

Non Spam Precision = 3/(2+3) = .6

	Label	Prediction	Outcome
1	spam	non-spam	
2	spam	spam	
3	non-spam	non-spam	
4	spam	spam	
5	non-spam	spam	
6	non-spam	non-spam	
7	spam	spam	
8	non-spam	spam	
9	non-spam	non-spam	
10	spam	spam	

Predicted Class

Spam	Non	Actual Class
4	2	Spam
1	3	Non

Precision vs. Recall

- No model will ever be perfect.
- Important to consider the performance required by the task.
  - Medical diagnosis → need a highly accurate diagnosis.
  - Predicting customer response to an ad → anything better than random will deliver profit.
  - Must align the evaluation measures with the modelling task.
- Precision or Recall?
  - Cancer diagnosis.
  - Recommendations from Netflix.
  - Stock market prediction.
  - Credit Card Fraud

F1 - Measure

- Precision and Recall can be collapsed into a single performance measure.
- F1-Measure, aka F1-score, is the harmonic mean of precision and recall.

$$\mathrm{F}_1\text{-measure} = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

Multiclass Performance

- Macro score - arithmetic mean of all per-class scores.
- Micro score - global score across all classes.
- Example:
  - Average Class Recall vs Accuracy

Imbalanced Data

- Imbalanced data: Refers to a problem in a classification where the classes in the data are not represented equally (i.e. the distribution of class sizes is skewed).
- **Example:** In a binary classification problem, we have a dataset of 100 items. 80 items belong to Class ***A***, and 20 belong to Class ***B***. This is an imbalanced dataset, where the ratio A:B is 4:1. We call ***A*** the majority class and ***B*** the minority class.
- This phenomenon occurs in many real-world problems:
  - Fraud detection: The vast majority of financial transactions are legitimate, and a small minority are fraudulent.
  - Churn analysis: Vast majority of customers stay with their mobile operator, a small minority cancel their subscription.
  - Other examples: medical diagnosis, e-commerce, security.

Example

- Consider the following example: In a binary classification problem, a dataset of 100 items. 90 items belong to Class A, 10 belong to Class B.
- Accuracy = 91% which seems good!
- The majority class overwhelms the performance of the model on the minority class.
- In imbalanced datasets, high accuracy can be achieved by just predicting the majority class

Predicted Class

A	B	
90	0	A
9	1	B

Actual Class

Balanced Measures

- To deal with skewed classes, use a balanced evaluation measure. Measures include:
  - Balanced Accuracy Rate (BAR): Mean of TPRate and TNRate aka Average Class Accuracy (ACA)
  - Balanced Error Rate (BER): Mean of FPRate and FNRate.
- For multi-class problems (macro score):

$$\text{ACA} = \frac{1}{|classes(t)|} \sum_{c \in classes(t)} \text{recall}_c$$

- ACA is an arithmetic mean which is susceptible to the influence of large outliers.

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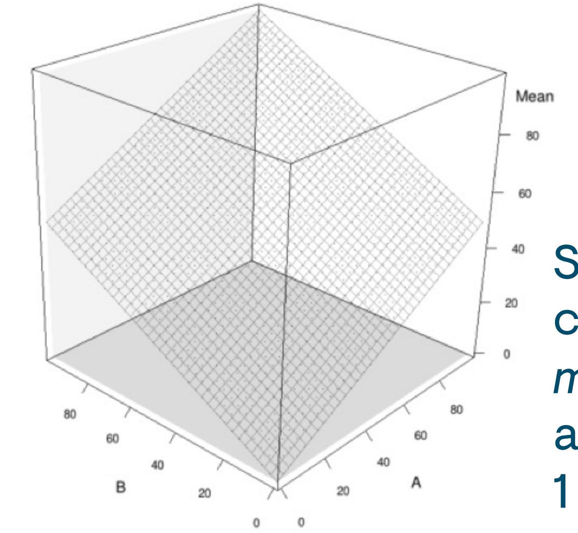
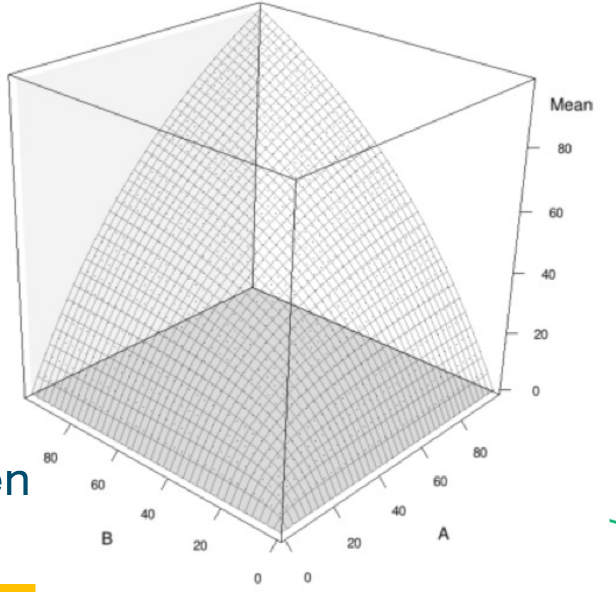
Balanced Measures

- Harmonic Mean mitigates the impact of large outliers and emphasises the importance of smaller values



(a)

Surfaces generated by calculating the (a) *arithmetic mean* and (b) *harmonic mean* of all combinations of A & B between 1 and 100



(b)

F1 Measure is a harmonic mean

Example

- Consider the following example: In a binary classification problem, a dataset of 100 items. 90 items belong to Class A, 10 belong to Class B.

Raw Accuracy = 91%
ACA = 55%
ACAHM = 18%

	Predicted Class A	Predicted Class B	
Actual Class A	90	0	A
Actual Class B	9	1	B

Prediction Score

- Binary classification models generally do not return a target class as the output, they return a prediction score that is converted to a class label using a threshold.
  - NB returns a value for each class, and we choose the largest.
  - DTrees returns the proportion of each class at a leaf node, and we choose the largest.
  - kNN returns the weighted distance of the nearest neighbours of each class, and we choose the largest.
- The prediction score is generally normalised in the range [0,1], and a threshold of 0.5 is used to convert this score to a class.

If score is greater than 0.5, assign to positive class else assign to negative class

This is effectively the majority / largest bit

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Previous example from naïve bayes

ID	HEADACHE	FEVER	VOMITING	MENINGITIS
1	true	true	false	false
2	false	true	false	false
3	true	false	true	false
4	true	false	true	false
5	false	true	false	true
6	true	false	true	false
7	true	false	true	false
8	true	false	true	true
9	false	true	false	false
10	true	false	true	true

What is the diagnosis for someone with a headache, fever but no vomiting?

$$\arg\max_{c \in classes(t)} \prod_{i=1}^{n} P(q_i \mid t = c) \times P(t = c)$$

Two classes: $m = T$ & $m = F$

$m=T$: $P(h=T \mid m=T) \times P(f=T \mid m=T) \times P(v=F \mid m=T) \times P(m=T)$
$= 2/3 \times 1/3 \times 1/3 \times 3/10 = 0.0148$

$m=F$: $P(h=T \mid m=F) \times P(f=T \mid m=F) \times P(v=F \mid m=F) \times P(m=F)$
$= 5/7 \times 3/7 \times 3/7 \times 7/10 = 0.0918$

$$\text{score}(m = T) = \frac{0.0148}{0.0148 + 0.0918} = 0.14 \qquad \text{score}(m = F) = \frac{0.0918}{0.0148 + 0.0918} = 0.86$$

Score threshold

- In some applications, the cost of a false positive or false negative is very high, e.g., clinical diagnosis or fraud detection.
- The threshold is used to discriminate when selecting between a positive and negative outcome.
- The threshold used on the prediction score can be moved to change the sensitivity of the model.
- Varying the decision threshold value can lead to different results, and so to different confusion matrices.

ROC Analysis

ID	Target	Pred-iction	Score	Out-come
7	ham	ham	0.001	TN
11	ham	ham	0.003	TN
15	ham	ham	0.059	TN
13	ham	ham	0.064	TN
19	ham	ham	0.094	TN
12	spam	ham	0.160	FN
2	spam	ham	0.184	FN
3	ham	ham	0.226	TN
16	ham	ham	0.246	TN
1	spam	~~ham~~	0.293	~~FN~~ TP

0.25

ID	Target	Pred-iction	Score	Out-come
5	ham	~~ham~~	0.302	~~TN~~ FP
14	ham	~~ham~~	0.348	~~TN~~ FP
17	ham	~~spam~~	0.657	~~FP~~ TN
8	spam	~~spam~~	0.676	~~TP~~ FN
6	spam	~~spam~~	0.719	~~TP~~ FN
10	spam	spam	0.781	TP
18	spam	spam	0.833	TP
20	ham	spam	0.877	FP
9	spam	spam	0.960	TP
4	spam	spam	0.963	TP

0.5

0.75

Threshold 0.75

Predicted Class: Ham	Spam	Actual Class
TN=10	FP=1	Ham
FN=5	TP=4	Spam

Threshold 0.5

Predicted Class: Ham	Spam	Actual Class
TN=9	FP=2	Ham
FN=3	TP=6	Spam

Threshold 0.2

Predicted Class: Ham	Spam	Actual Class
TN=7	FP=4	Ham
FN=2	TP=7	Spam

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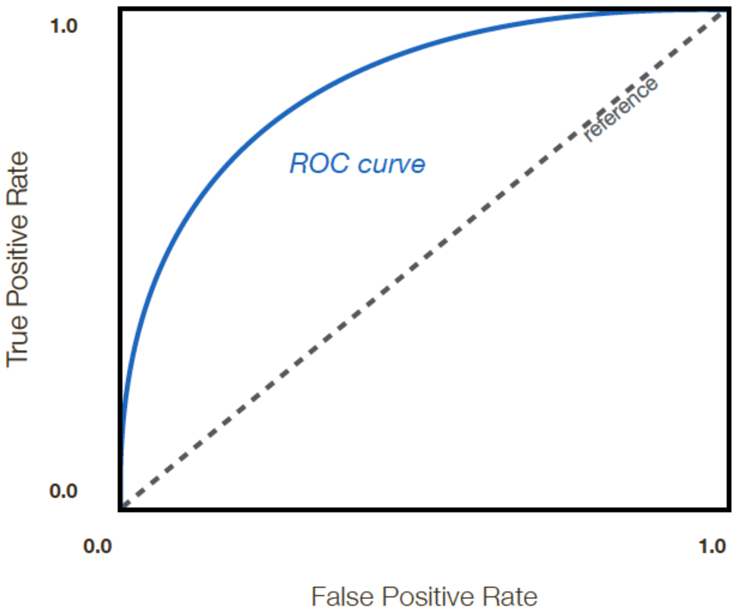
ROC analysis

- TPR and TNR are intrinsically tied to the threshold used for the prediction score.
- There is a trade off between TPR and TNR - the Receiver Operating Characteristic (ROC) Curve captures this trade off.
- Plot TPR on vertical axis against 1-TNR (= FPR) on the horizontal axis for a range of score threshold values.
- A trained classifier should always be above the 'random' reference line.



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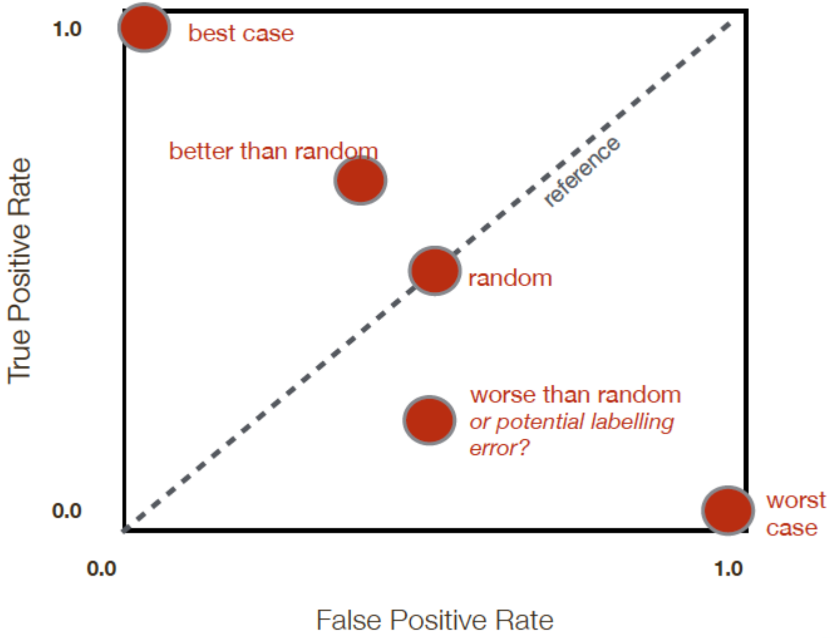
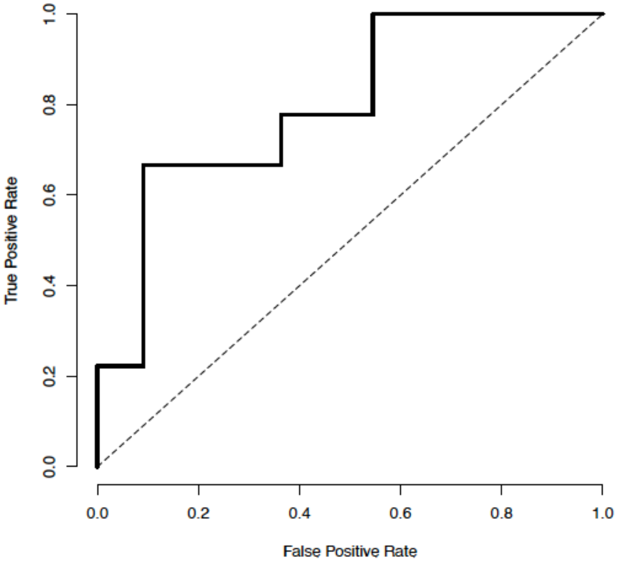
ROC analysis

- As strength of the model increases the ROC moves to top left hand corner (TPR =1; TNR=1).

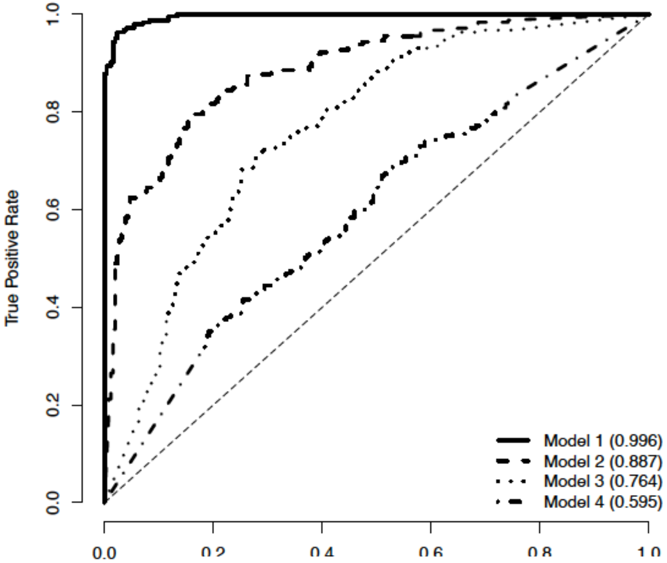


ROC Analysis

- Often ROC curves for multiple prediction models are plotted on a single ROC plot, showing easy comparison of performance.



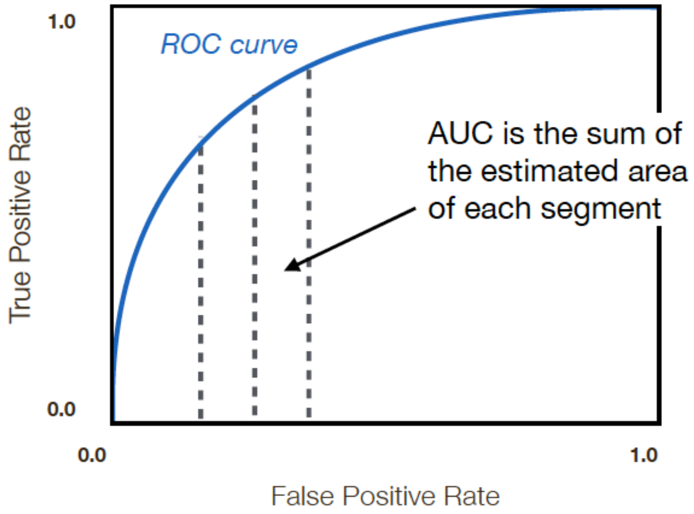
ROC curve for spam example dataset

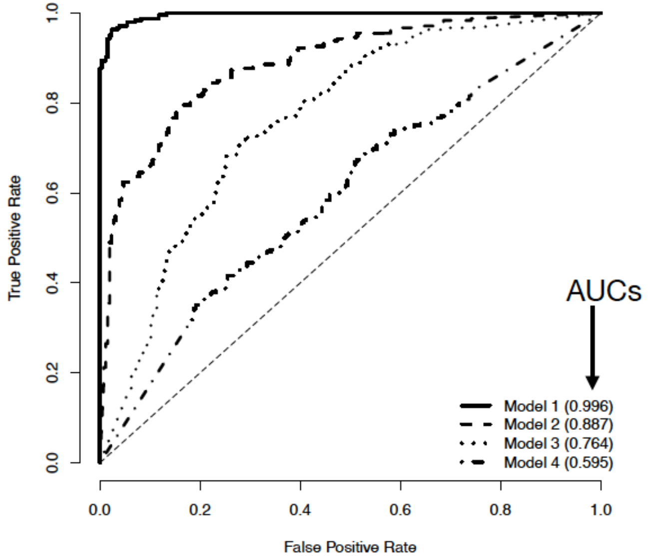


ROC curves for 4 models on a significantly larger spam dataset

AUC

- Comparing ROC curves visually is useful, but it is preferable to have a single measure to assess models.
- Area Under the ROC Curve (AUC) measures the area beneath an ROC curve, a bigger area is better performance.
- Calculated as the integral of the ROC curve but can be done using the trapezoidal method





Performance measures – continuous targets

- Sum of Squared Errors (SSE)
  - Most common measure for continuous targets

$$\mathrm{SSE} = \frac{1}{2}\sum_{i=1}^{n}\left(t_i - \hat{t_i}\right)^2$$

t_i is target value of instance i

$\hat{t_i}$ is predicted value of instance i

- Mean Squared Error (MSE)
  - Average difference between predictions and actual values of the test set.
  - Allows the ranking of performance of multiple models.
  - Not meaningful in relation to the scenario the model is being used for (error not in the same units as target).

$$\mathrm{MSE} = \frac{\sum_{i=1}^{n}\left(t_i - \hat{t_i}\right)^2}{n}$$

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Performance measures – continuous targets

- Root Mean Squared Error (RMSE)
  - in the same units as the target value so more meaningful
  - Tends to over emphasise individual large errors (due to square)

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} \left(t_i - \hat{t_i}\right)^2}{n}}$$

- Mean Absolute Error (MAE)
  - addresses the problem of using the square (similar to using Manhattan over Euclidean distance)

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n} |t_i - \hat{t_i}|}{n}$$

- But both require domain knowledge…

Performance measures – continuous targets

- R^2Coefficient is a normalised domain independent measure of model performance.

$$R^2 = 1 - \frac{\text{sum of squared errors}}{\text{total sum of squares}}$$

$$\text{total sum of squares} = \frac{1}{2}\sum_{i=1}^{n}\left(t_i - \bar{t}\right)^2$$

- Compares the performance of a model on a test dataset with the performance of an imaginary model that always predicts the average values from the test dataset
  - always in range [0,1]
  - higher numbers better

Example

- Comparing the prediction of two models for a blood-thinning drug dosage prediction problem.

		Linear Regression		k-NN	
ID	Target	Prediction	Error	Prediction	Error
1	10.502	10.730	0.228	12.240	1.738
2	18.990	17.578	-1.412	21.000	2.010
3	20.000	21.760	1.760	16.973	-3.027
4	6.883	7.001	0.118	7.543	0.660
5	5.351	5.244	-0.107	8.383	3.032
6	11.120	10.842	-0.278	10.228	-0.892
...					
30	3.538	7.090	3.551	5.553	2.014
	MSE		**1.905**		**4.394**
	RMSE		**1.380**		**2.096**
	MAE		**0.975**		**1.750**
	R^2		**0.889**		**0.776**

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What measure to use?

- Guideline:

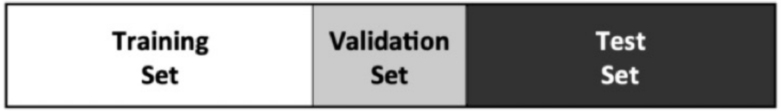
  Better to be pessimistic when evaluating performance

- Classification
  - Recommend use of Average Class Accuracy using harmonic mean
- Regression (continuous targets)
  - Recommend use of R^2Coefficient

Designing evaluation experiments

- Distinct, random, non-overlapping samples of data needed for evaluation
- Hold-out sampling divides the dataset into training, validation and test sets
  - No fixed recommendations on proportions, 50:20:30 or 40:20:40 are common.
- Issues:
  - Is there enough data?
  - Can be misleading with ‘lucky’ splits of the data

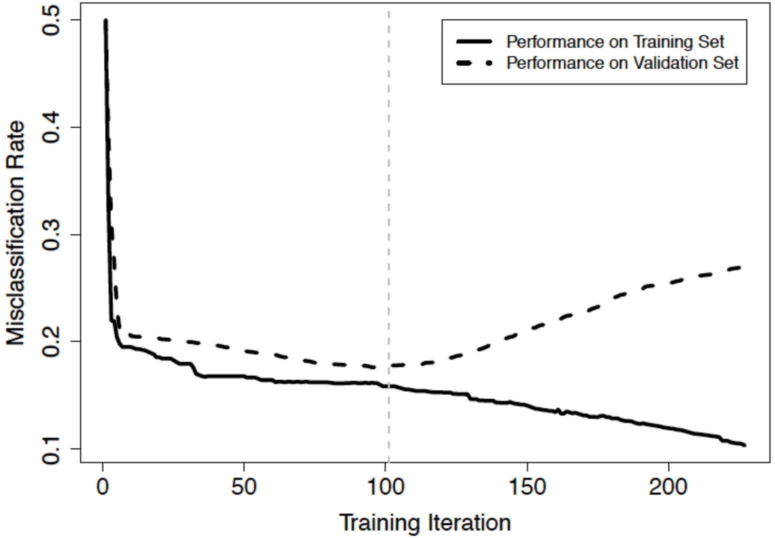
(a) A 50:20:30 split



(b) A 40:20:40 split

Validation set

- Validation set is used for tuning the model, e.g. hyperparameter fixing or feature selection.
- Also useful in iterative ML algorithms, e.g. decision trees, regression models. to avoid overfitting



Similar to post-pruning process in decision trees

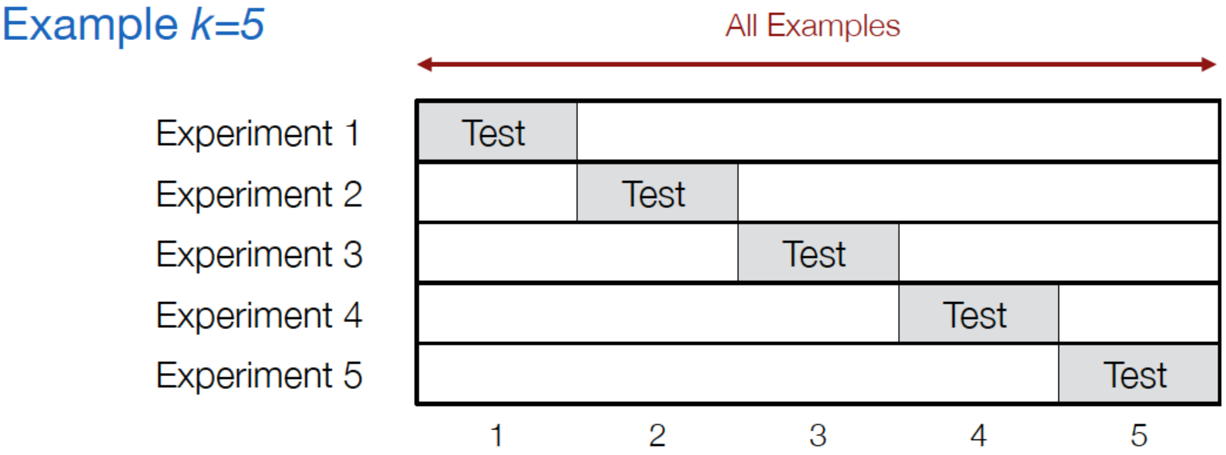
Recap on overfitting

- For real-world tasks, we are interested in generalisation accuracy.
- Overfitting: Model is fitted too closely to the training data (including its noise). The model cannot generalise to situations not presented during training, so it is not useful when applied to unseen data.
- **Possible Causes**
  - Small training set: Classifier only given a few examples, may not be representative of the underlying concepts.
  - Complex model: Model has too many parameters relative to the number of training examples.
  - Noise: Spurious or contradictory patterns in the training data.
  - High-dimensionality: Data has many irrelevant features (dimensions) containing noise which leads to a poor model.

➡️ A good model must not only fit the training data well, but also accurately classify examples that it has never seen before.

K-fold Cross validation

- k-Fold Cross Validation
- Divide the data into k disjoint subsets - “folds” (e.g. k=5 or 10).
- For each of k experiments, use k-1 folds for training and the selected one fold for testing.
- Repeat for all k folds, average the accuracy/error rates.



Example: Cross-validation

- Number of correct and incorrect predictions made by a spam classifier on 300 emails, when we run 5-fold cross validation.

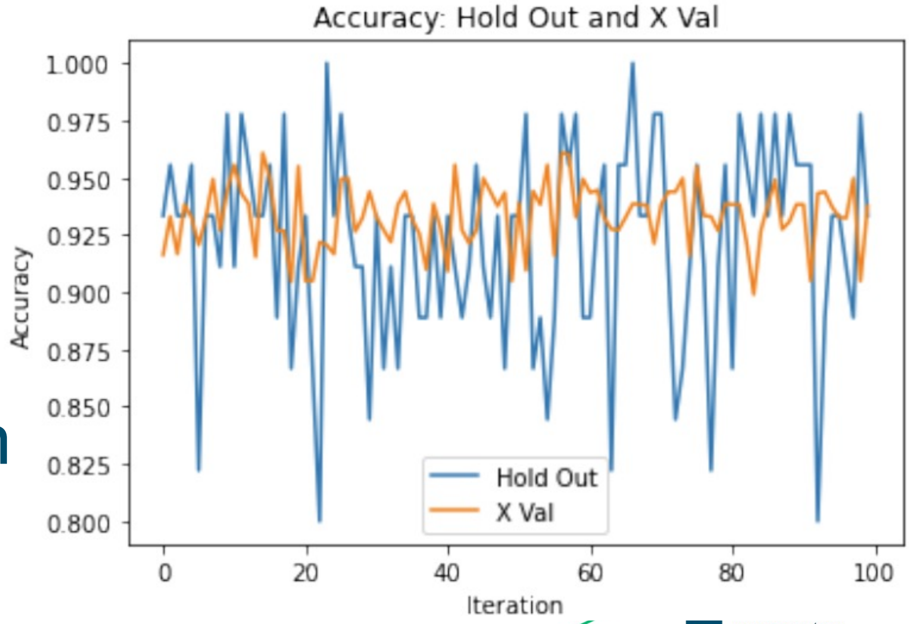
	Class: Non-Spam		Class: Spam		
Fold	Correct	Incorrect	Correct	Incorrect	**Accuracy**
1	173	22	87	18	86.67%
2	107	88	71	34	59.33%
3	143	52	80	25	74.33%
4	185	10	59	46	81.33%
5	162	33	71	34	77.67%
Mean	154	41	74	31	**75.87%**

Accuracy for each fold

Average accuracy across all 5 folds

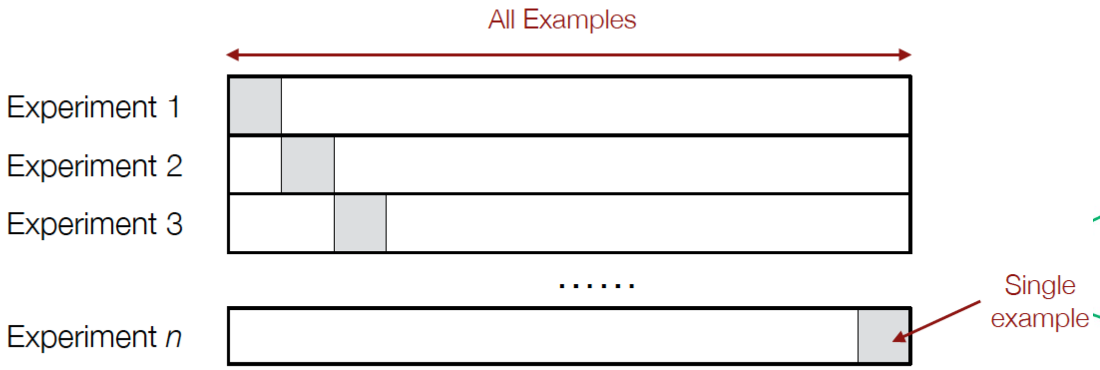
Holdout vs. cross-validation

- Cross Validation
  - ✓ More stable estimate of accuracy
  - ✓ Better estimate of accuracy (Hold-Out can underestimate)
  - Slower ✗



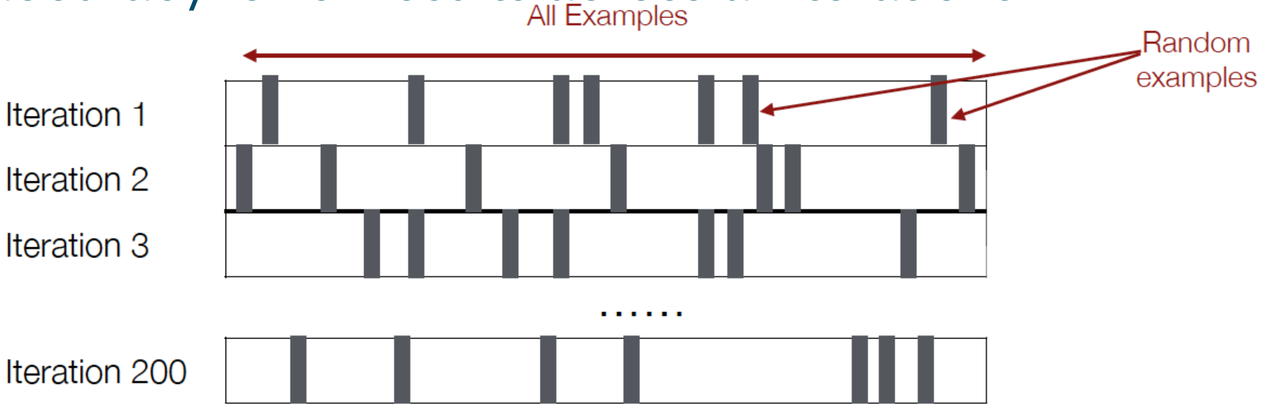
Leave-one-out Cross Validation

- Leave-one-out (LOO) Extreme case of k-fold cross-validation where k is selected to be the total number of examples in the dataset.
- For a dataset with n examples, perform n experiments.
- For each experiment, use n-1 examples for training and the remaining single example for testing.
- Average the accuracy/error rates over all n experiments.



Bootstrapping

- Bootstrapping approaches are preferred over cross-validation in contexts with very small datasets (<300 examples)
- A random selection of examples is selected for testing, remaining used for training
- Repeat multiple times, generally significantly more than *k* in *k-fold* CV (>=200 iterations)
- Average accuracy/error results across all iterations



Summary

- Choice of performance measures depends on
  A. nature of the problem, i.e. continuous vs categorical;
  B. characteristics of the dataset, i.e. balanced vs unbalanced;
  C. needs of the application, (e.g. medical diagnosis vs marketing response prediction) or domain
- Recommend in absence of other factors
  – Classification - use ACA_{HM}
  – Regression – use R^2
- Data splitting approach
  – Very small datasets, use Bootstrapping
  – Very large datasets, use Holdout sampling (average over n iterations)
  – Otherwise, consider Cross-Validation
  – May need to consider natural structure in the data, e.g. where there is a time dimension, use data from one period for training and another period for testing (known as out-of-time sampling)

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Questions?